THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
KASULU TOWN FORM II-MOCK EXAMINATION 2019

CODE:041 **BASIC MATHEMATICS**

TIME 2:30 HOURS

INSTRUCTION

1. This paper consists of ten (10) compulsory questions.
2. Show clearly all the working and answers in the space provided.
3. All writtings must be in blue or black ink except drawings which must be in pencil.
4. Four figure/mathematical tables(NECTA) geometric instruments and graph papers may be used where necessary.
5. All communication devices,calculators and any unauthorised materials are not allowed in the examination room.
6. Write your **examination number** at the top right corner of every page.

FOR EXAMINER'S USE ONLY

QUESTION NUMBER	SCORE	EXAMINER'S INITIALS
01		
02		
03		
04		
05		
06		
07		
08		
09		
10		
TOTAL		
ENTERER'S INITIALS		
CHECKER'INITIALS		

1. (a) Find fractional notation for $0.\dot{8}3\dot{5}$

1. (b) Mariam was given 20,000/= by her father. She spent 48% of it to buy shoes. How much money remained?

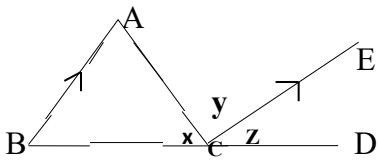
2. (a) Evaluate 0.3143 by 6.06 giving the answer correct to 3 significant figures.

(b) Arrange in increasing order after converting each of the following into metres.
68hm , 0.68km and 16800cm

3. (a) Round off 0.597 to 2 significant figures.

3. (b) Study carefully the figure below where $< BAC = 50° and° < ABC = 70°$, then find the value of

x, y and z



4. (a) solve the following inequality $\frac{2}{3}x - 1 \leq 3x - 7$

(b) Find the principal that will earn Shs. 72,900 at the rate of $2\frac{1}{2}\%$ per annum in 8 years.

5. (a) Find the equation of a line containing the following pairs of points: (7,9), (2,5)

(b) the length of a rectangle is twice its width, and its area is 20,000cm^2. What is the width?

6. (a) Express the number $\frac{2+\sqrt{3}}{\sqrt{2}-\sqrt{5}}$ in the form of $\frac{\sqrt{a}}{b}$

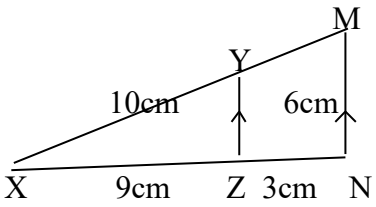
(b) Solve the equation $x^2 + 5x - 14 = 0$ by completing the square.

7. (a) Determine y from $\log(y^2 + 3y - 44) = 1$

(b) The figure below shows that AC=BD. Prove that ACB=ADB.

8. (a) From the figure below triangles XYZ and XMN are equingular and similar. Calculate MN and

MY.



(b) By using a sketch, find the image of the point A(5,2) under a reflection in the line y=0, followed by another reflection in the line y=x.

9. (a) A ladder on the ground leans against a vertical wall whose height is 5metres. The ground distance between the ladder and the wall is 12metres.Draw a diagram to represent this information and use it to find the length of the ladder.

(b) Given that sinA=$^{3}/_{5}$ where A is an acute angle, find without using mathematical tables

the value of $\frac{1-sinA}{1-cosA}$

10. (a) In a class of 32 students, 18 play golf, 16 play piano and 7 play both golf and piano.Use a formula to find the number of students who play neither golf nor piano.

(b) Show this information in a pie chart.

Occupations	Doctors	Lawyers	Engineers	Pilots	Geologists	Farmers
Number	25	40	60	20	40	15

THE END